

Sinan-Deniz Ceviker *AI & Software Engineer*

Groningen, NL | denizceviker12@gmail.com | LinkedIn | GitHub | Website

Hands-on engineer building production-ready systems across machine learning, full-stack web, mobile, and desktop apps.

PROFESSIONAL EXPERIENCE

- Software Engineer Internship**, TNO 02/2026 – 06/2026
- Implement backend services exposing standardized access to domain-specific data models for AI systems.
 - Prototype AI interoperability protocols and developer tooling, mapping heterogeneous APIs into unified MCP-based interfaces.
- Machine Learning Intern**, Youwe 09/2025 – 02/2026
- Built a **Sprint Health Classifier** using synthetic and real Agile data, reconstructing sprint states over time via preprocessing of Jira exports.
 - Designed labeling logic and trained models robust to distribution shift, achieving **96% accuracy on unhealthy sprints from sprint start** and near-perfect classification at completion.
- Frontend Developer (Part-Time)**, University of Groningen 08/2025 – Present
- Built interactive dashboards for building occupancy and capacity planning, working with university stakeholders through **scheduled demos and feedback cycles**.
 - Implemented performant data visualizations supporting energy efficiency and sustainability initiatives.
- Software Engineer**, Freelancer 08/2024 – Present
- Delivered end-to-end software systems (web, mobile, desktop) for SMEs across Europe.
 - Built automation-driven platforms with Next.js, NestJS, React Native, Expo, and Tauri.

PROJECTS

- AI Cup 2026 - Bird Species Classification from Radar Data**, 02/2026 – 03/2026
(2nd place / 89 teams)
- End-to-end ML pipeline for 9-class classification; hosted by Team Epoch (TU Delft) with TNO as partner, and evaluated on macro-averaged Average Precision.
 - Architected solution using TabPFN as core classifier with TabEBM generated synthetic samples for rare-class augmentation. Injected domain priors derived from external bird migration data to bias predictions towards biologically plausible distributions.
 - Feature engineering, including residualization conditioned on weather parameters via grid search to disentangle biological signal from meteorological variation & temporal shift; followed by feature pruning based on inter-class discriminativity.
- PLC Remote Control App**, Industrial PLC + BLE
- Designed a **machine-agnostic control engine** using a schema-driven controller abstraction, enabling runtime support for multiple PLC models without code changes.
 - Built an **adapter-based architecture** decoupling transport, protocol framing, and machine state, with a serialized command queue, inter-command timing guarantees, and optimistic state handling.
 - Implemented PLC communication (checksum validation, mixed register reads, discrete controls) and shipped a **production Android APK** used in-field, replacing physical HMIs and reducing operational friction and cost.

EDUCATION

- Bachelor's Degree, Artificial Intelligence**, University of Groningen 09/2023 – 2026
- **Specialization:** Machine Learning (Neural Networks, Reinforcement Learning, Uncertainty in ML) and Robotics (Robotics Practical, Social Robotics)
 - Coursework highlights: Applied Machine Learning, Robotics Practical, Signals & Systems, Data Science

SKILLS

Software Engineering: Next.js, React, TypeScript, Node.js, NestJS, React Native, Tauri, Rust

AI: Python, Machine Learning, Neural Networks, Reinforcement Learning, Time Series Forecasting, Anomaly Detection, RAG

Systems & Data: Docker, Linux, CI/CD, SQL, NoSQL, API Design, Data Pipelines, REST APIs

Embedded: PLC, Bluetooth Low Energy (BLE), Embedded Systems, C, QNX